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Reliability optimization of weighted k-out-of-n:G systems under dual fuzziness

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Abstract

A weighted k-out-of-n:G system works when the total capacity of its working components reaches a demand threshold. In early-stage design both the unit capacities and the demand are often expert estimates, yet the weighted models with imprecise capacities that we identified keep the demand crisp and evaluate given designs. This paper models capacities and demand as triangular fuzzy numbers and compares them by a Hurwicz rule on α-cuts with a cut level and an optimism index. The rule contains the crisp model, the interval worst and best cases and the necessity-, possibility- and credibility-based chance constraints, so the attitude towards imprecision becomes an explicit design input. When all capacities share one proportional spread, the two indices act on the design only through one integer threshold; a two-phase branch-and-bound algorithm certifies the optimal design at each threshold, and a finite list of designs covers every index value. On a generation fleet built from the IEEE Reliability Test System, seven fleets costing \$130–203 M cover the whole index range, two of which a 30-cell grid over the indices never finds, and the crisp design fails the reliability target on 6.8% of the range and overpays on a further 32.1%.

Keywords

Weighted k-out-of-n:G system, fuzzy reliability, redundancy allocation, possibilistic chance constraint, Hurwicz criterion, branch-and-bound, generation adequacy

1 Introduction

A weighted k-out-of-n:G system (Wk/n:G) consists of n independent components, component i contributes an operational capacity $w_i > 0$, and the system works if and only if the total capacity of the working components reaches a threshold k:

$$\sum_{i=1}^{n} w_i \, s_i \;\geq\; k, \quad s_i \in \{0, 1\},$$

where $s_i = 1$ when component i works. The model generalizes the k-out-of-n:G system to components of unequal performance, as in generation fleets, pumping stations and production lines with parallel units. Wu and Chen[1] gave the $O(n \cdot k)$ recursion that underlies later evaluation methods[2]. The model has since been extended to multi-state components and performance levels[3–5], three-state components[6], randomly weighted and heterogeneous components[7,8], the capacity lost upon system failure[9], dynamic and time-dependent settings of weighted and related k-out-of-n systems[10,11], several component types with optimization of the design[12], and maintenance planning[13–16]; related k-out-of-n structures continue to be generalized[17].

Most of this literature treats the capacities w_i as precisely known, and the threshold k as a fixed constant. In early-stage infrastructure design neither holds. Unit capacities change with wear and ambient conditions and are often known through expert estimates, and the threshold is a design judgement derived from regulations, demand forecasts or engineering consensus. Both carry epistemic uncertainty, which arises from incomplete knowledge rather than from variability in repeated trials, and for which a probability distribution adds an assumption that the data do not support[18,19]; possibility theory represents such knowledge without a distributional assumption[20,21]. To make the consequences concrete, consider a regional grid operator that must deliver "at least 180 MW" at peak demand. The figure is a regulator's target negotiated from demand

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projections, and the available output of each unit depends on ambient temperature and wear. A crisp formulation fixes both sides at nominal values and thereby adopts a risk posture without stating it: a design feasible at nominal capacities may have zero necessity of meeting demand over the plausible range of capacities, whereas requiring demand to be met with high necessity may need considerably more redundancy and capital.

Imprecise thresholds and capacities. Several threshold models relax the crisp k, but each keeps the threshold precise in some sense: multi-state models use several crisp values[3], phased-mission models a crisp value per phase[22,23], and random-threshold models a probability distribution that needs frequency data[24]. On the component side, Ding et al.[4] represent the weights and state probabilities of multi-state weighted systems by fuzzy numbers and evaluate adequacy against a prescribed demand; within Liu's uncertainty theory, Gao et al.[25] and Wei et al.[26,27] analyze weighted systems whose weights and states are uncertain variables, and Sheng and Ke[28] multi-state k-out-of-n systems. In all of these the threshold is a prescribed constant, and no design problem is posed. Fuzzy reliability expressions for weighted and consecutive structures[29–31], possibilistic reliability of k-out-of-n systems[32] and fuzzy or possibilistic allocation models[33–36] likewise fuzzify the component side while the failure boundary stays crisp; in a fuzzy-goal model of a multi-state weighted allocation the reliability requirement and the budget are fuzzy, but capacities and threshold are crisp[37]. Allocation models also fix one reading of the imprecise data in advance: a possibility[38] or credibility[39,40] chance constraint, or a minimax-regret criterion over intervals[41].

Once both sides of the success condition are fuzzy, the condition is no longer an inequality between numbers, and a comparison rule must be chosen. The rule decides which subsets of working components count as successful and therefore what the reported reliability means, yet it is rarely stated. Table 1 compares this work with the closest studies.

Contributions. This paper poses the redundancy-allocation problem of a Wk/n:G system whose capacities and demand are both triangular fuzzy numbers and solves it exactly. It makes three contributions.

1. **A declared comparison rule.** Capacity and demand are compared by a Hurwicz rule on α-cuts with two parameters, a cut level α and an optimism index γ. The usual readings of imprecise data (the crisp model, the interval worst and best cases, and necessity-, possibility-

Table 1. Comparison with the closest related Wk/n:G frameworks. "Uncertain": Liu's uncertain measure rather than a possibility distribution. "n/a": no comparison between two imprecise quantities arises. Design task: "Eval." computes the reliability of a given design; "Heuristic opt." optimizes the design without a certificate of global optimality; "Exact opt." optimizes it and certifies the global optimum.

Feature	Ding et al.[4]	Gao et al.[25]	Negi et al.[30]	Ashraf et al.[34]	Eryilmaz & Ucum[9]	Maneckshaw et al.[36]	Wei et al.[26]	**This paper**
$\tilde{w}_i$ Fuzzy	Yes	Uncertain	No	No	No	No	Uncertain	**Yes**
$\tilde{k}$ Fuzzy	No	No	No	No	No	No	No	**Yes**
Design task	Eval.	Eval.	Eval.	Heuristic opt.	Eval.	Heuristic opt.	Eval.	**Exact opt.**
Declared ordering	No	n/a	No	No	n/a	No	n/a	**Yes**

and credibility-based chance constraints) are points of this policy square, so they can be compared on one instance with one solver.

2. **The optimal design for every parameter value.** Because the system succeeds through the sum of its capacities, a common proportional spread scales that sum by one factor, and the two parameters act on the design only through one integer threshold. A number of crisp problems of the same order as a 30-cell grid, one per threshold, gives the optimal design at every point of the square instead of at the points of a grid, together with the exact regions on which each design is optimal.
3. **A certified exact solver and an engineering case study.** A two-phase branch-and-bound algorithm returns the most reliable of the minimum-cost designs and certifies its global optimality. On a generation fleet built from the IEEE Reliability Test System (RTS-79)[42], the complete solution is a short menu of fleets with their cost and guarantee, and it shows what a grid study, the crisp model, one-sided fuzzy models and a probabilistic counterpart lose.

The model and the rule are developed in Section 2, the design problem and its exact solution in Section 3, the case study in Section 4 and the results in Section 5; Section 6 discusses how a posture is chosen and the limitations. Proofs, the notation table, two literature-derived instances that replicate the findings, the scaling study and the script-level provenance of every number are given in the Supplementary Material.

2 Dual-fuzzy model and comparison rule

2.1 System model and α-cuts

The full notation is listed in Supplementary Material S11. Component $i \in \{1, \ldots, n\}$ has state $s_i \in \{0, 1\}$ and capacity $w_i > 0$; for the set $S = \{i : s_i = 1\}$ of working components the modal total capacity is $W_S = \sum_{i \in S} w_i$, and the crisp system works if $W_S \geq k$. When capacities and demand are imprecise, the dual-fuzzy model replaces them by triangular fuzzy numbers (TFNs) $\tilde{w}_i = (w_{i,L}^0, w_i, w_{i,U}^0)$ and $\tilde{k} = (k_1, k_2, k_3)$. By Zadeh's extension principle the aggregate capacity of S is $\tilde{W}_S = \sum_{i \in S} \tilde{w}_i$, and the success condition becomes the comparison of two fuzzy quantities, $\tilde{W}_S \tilde{\geq} \tilde{k}$. The α-cut of a TFN $\tilde{A} = (a_1, a_2, a_3)$ at cut level $\alpha \in [0, 1]$ is the interval

$$[A_L^\alpha, A_U^\alpha] = [a_1 + \alpha(a_2 - a_1),\ a_3 - \alpha(a_3 - a_2)], \tag{1}$$

the closure of the support at $\alpha = 0$; the cuts are nested and shrink to the modal value at $\alpha = 1$.

2.2 Hurwicz evaluation rule

At cut level α the aggregate capacity of S is the interval $[W_{S,L}^\alpha, W_{S,U}^\alpha]$ and the threshold is $[k_L^\alpha, k_U^\alpha]$. When the two intervals overlap, the capacity is neither surely above nor surely below the demand, and the success condition is undefined until a comparison rule is chosen. We compare the Hurwicz values of the two intervals[43], weighted by an optimism index $\gamma \in [0, 1]$:

$$W_S^\alpha(\gamma) = \gamma W_{S,U}^\alpha + (1 - \gamma) W_{S,L}^\alpha, \tag{2}$$

$$k^\alpha(\gamma) = k_U^\alpha - \gamma(k_U^\alpha - k_L^\alpha). \tag{3}$$

Definition 1 (Hurwicz evaluation rule). For $\alpha, \gamma \in [0, 1]$, the fuzzy success condition $\tilde{W}_S \tilde{\geq} \tilde{k}$ holds at (α, γ) if and only if

$$W_S^\alpha(\gamma) \geq k^\alpha(\gamma). \tag{4}$$

At $\gamma = 0$ the rule compares the lowest capacity with the highest threshold (pessimistic evaluation); at $\gamma = 1$ the highest capacity with the lowest threshold (optimistic evaluation). The convex weighting of interval endpoints is used for ranking fuzzy numbers under different risk attitudes[44–46]. The rule is defined on α-cut intervals rather than on

event spaces[47]; Section 2.3 shows that its extreme columns nevertheless coincide with necessity- and possibility-based chance constraints.

Lemma 1 (Crisp equivalence). *For any subset S, TFN capacities and demand, and $\alpha, \gamma \in [0,1]$, let $w_i^\alpha(\gamma) = \gamma\, w_{i,U}^\alpha + (1-\gamma)\, w_{i,L}^\alpha$. Then Equation* (4) *holds if and only if*

$$\sum_{i \in S} w_i^\alpha(\gamma) \;\geq\; k^\alpha(\gamma). \tag{5}$$

Proof. The α-cut endpoints of a sum of fuzzy numbers are the sums of the endpoints, so $W_S^\alpha(\gamma) = \sum_{i\in S}[\gamma w_{i,U}^\alpha + (1-\gamma) w_{i,L}^\alpha]$. □

For a fixed pair (α, γ), with common or unit-specific spreads, the fuzzy success condition is therefore a crisp Wk/n:G condition with capacities $w_i^\alpha(\gamma)$ and threshold $k^\alpha(\gamma)$, and dual fuzziness defines a parameterized family of crisp Wk/n:G problems.

2.3 Possibilistic readings of the rule

For TFNs $\tilde{A} = (a_1, a_2, a_3)$ and $\tilde{B} = (b_1, b_2, b_3)$ the possibility and necessity of the event $\tilde{A} \geq \tilde{B}$ are[48]

$$\Pi(\tilde{A} \geq \tilde{B}) = \begin{cases} 1, & a_2 \geq b_2, \\ \dfrac{a_3 - b_1}{(a_3 - a_2) + (b_2 - b_1)}, & \text{otherwise}, \end{cases} \tag{6}$$

$$N(\tilde{A} \geq \tilde{B}) = \begin{cases} 0, & a_2 \leq b_2, \\ 1 - \dfrac{b_3 - a_1}{(b_3 - b_2) + (a_2 - a_1)}, & \text{otherwise}, \end{cases} \tag{7}$$

both clipped to $[0,1]$. Possibility is the optimistic reading and necessity the pessimistic one; the credibility measure $\mathrm{Cr} = \frac{1}{2}(\Pi + N)$ is self-dual[39]. A decision maker may fix a confidence level λ instead of (α, γ) and require $N(\tilde{W}_S \geq \tilde{k}) \geq \lambda$, a possibilistic chance constraint[20]. Proposition 1 maps such constraints onto cells of the policy square; the proofs of the propositions and corollaries of this paper are given in Supplementary Material S10.

Proposition 1 (Chance-constraint correspondence). *Let the capacities $\tilde{w}_i$ and the threshold $\tilde{k}$ be TFNs, and let the spreads that govern each branch be non-degenerate: $\sum_{i\in S}(w_i - w_{i,L}^0) + (k_3 - k_2) > 0$ for the necessity direction and $\sum_{i\in S}(w_{i,U}^0 - w_i) + (k_2 - k_1) > 0$ for the*

possibility direction. Then, for every confidence level $\lambda \in (0, 1]$*,*

$$N(\tilde{W}_S \geq \tilde{k}) \geq \lambda \iff W_S^{\alpha}(\gamma) \geq k^{\alpha}(\gamma) \quad \text{at } (\alpha, \gamma) = (1 - \lambda,\, 0), \tag{8}$$

$$\Pi(\tilde{W}_S \geq \tilde{k}) \geq \lambda \iff W_S^{\alpha}(\gamma) \geq k^{\alpha}(\gamma) \quad \text{at } (\alpha, \gamma) = (\lambda,\, 1). \tag{9}$$

Corollary 1 (Credibility chance constraints). *Under the assumptions of Proposition 1, for every* $\lambda \in (0, 1]$*,* $\mathrm{Cr}(\tilde{W}_S \geq \tilde{k}) \geq \lambda$ *holds if and only if Equation* (4) *holds at*

$$(\alpha, \gamma) = \begin{cases} (2\lambda,\, 1), & \lambda \leq 1/2, \\ (2 - 2\lambda,\, 0), & \lambda > 1/2. \end{cases} \tag{10}$$

In the notation of the policy square, the equivalences restate the standard endpoint forms of necessity and possibility constraints[20,49–51]: as α sweeps $[0, 1]$, the column $\gamma = 0$ enumerates every necessity-based chance constraint and the column $\gamma = 1$ every possibility-based one. A requirement such as "demand met with necessity at least 0.7" is the cell $(0.3, 0)$. The interior columns interpolate between the two attitudes, and with symmetric data the central column $\gamma = 1/2$ is the modal comparison $W_S \geq k_2$, which is also the credibility constraint $\mathrm{Cr} \geq 1/2$. Hurwicz evaluation and possibilistic chance-constrained programming are therefore handled by one solver. Table 2 collects the readings that the square contains.

Table 2. Classical readings of the success condition $\tilde{W}_S \geq \tilde{k}$ as points of the policy square (triangular capacities and threshold; Proposition 1 and Corollary 1, with $\lambda \in (0, 1]$). Readings marked "outside" are compared with the square in Section 5.3; Supplementary Material S6 maps each reading to its threshold and optimal design on the IEEE RTS-79 fleet.

Reading	Condition on subset S	Location in the square
Crisp model at the modal values	$W_S \geq k_2$	$\alpha = 1$, any γ
Interval worst case on the supports	$W^0_{S,L} \geq k^0_U$	$(0, 0)$
Interval best case on the supports	$W^0_{S,U} \geq k^0_L$	$(0, 1)$
Necessity chance constraint	$N(\tilde{W}_S \geq \tilde{k}) \geq \lambda$	$(1 - \lambda, 0)$
Possibility chance constraint	$\Pi(\tilde{W}_S \geq \tilde{k}) \geq \lambda$	$(\lambda, 1)$
Credibility chance constraint, $\lambda \leq 1/2$	$\mathrm{Cr}(\tilde{W}_S \geq \tilde{k}) \geq \lambda$	$(2\lambda, 1)$
Credibility chance constraint, $\lambda > 1/2$	$\mathrm{Cr}(\tilde{W}_S \geq \tilde{k}) \geq \lambda$	$(2 - 2\lambda, 0)$
Expected necessity and possibility reliability	R_N, R_Π (Eq. (12))	averages of the columns $\gamma = 0$, $\gamma = 1$
One-sided fuzzy models	$\tilde{w}_i$ or $\tilde{k}$ crisp	outside (other data)
Probabilistic counterpart	$\Pr\{\Theta W_S \geq D\}$	outside
Liu's uncertainty theory	uncertain measure	outside (Supplementary S4)

2.4 Reliability under each rule

A reliability measure aggregates over subsets. For a design $\mathbf{x}$ (Section 3.1) let $I(\mathbf{x})$ be the set of subsystems in which a unit is installed, p_i and $\tilde{w}_i$ the reliability and capacity of the installed unit, and $\Pr(S \mid \mathbf{x}) = \prod_{i \in S} p_i \prod_{l \in I(\mathbf{x}) \setminus S} (1 - p_l)$. The static reliability of $\mathbf{x}$ at (α, γ) is

$$R_{\mathrm{sys}}^{\alpha,\gamma}(\mathbf{x}) = \sum_{S \subseteq I(\mathbf{x}),\, \sum_{i \in S} w_i^{\alpha}(\gamma) \geq k^{\alpha}(\gamma)} \Pr(S \mid \mathbf{x}), \tag{11}$$

and the expected necessity and possibility reliabilities weight each subset by the necessity and the possibility of its success:

$$\begin{aligned} R_N(\mathbf{x}) &= \sum_{S \subseteq I(\mathbf{x})} \Pr(S \mid \mathbf{x})\, N(\tilde{W}_S \geq \tilde{k}), \\ R_{\Pi}(\mathbf{x}) &= \sum_{S \subseteq I(\mathbf{x})} \Pr(S \mid \mathbf{x})\, \Pi(\tilde{W}_S \geq \tilde{k}). \end{aligned} \tag{12}$$

Proposition 2 (Column averages). *If the assumptions of Proposition 1 hold for every subset $S \subseteq I(\mathbf{x})$ of a design $\mathbf{x}$, then*

$$R_N(\mathbf{x}) \;=\; \int_0^1 R_{\mathrm{sys}}^{\alpha,0}(\mathbf{x})\, d\alpha, \qquad R_{\Pi}(\mathbf{x}) \;=\; \int_0^1 R_{\mathrm{sys}}^{\alpha,1}(\mathbf{x})\, d\alpha. \tag{13}$$

The expected necessity and possibility reliabilities are thus the averages of the pessimistic and optimistic columns over the cut level. A single cell also has a chance-constrained reading: by Proposition 1, $R_{\mathrm{sys}}^{1-\lambda,0}(\mathbf{x}) \geq R_{\mathrm{target}}$ is equivalent to $\Pr\{N(\tilde{W}_S \geq \tilde{k}) \geq \lambda\} \geq R_{\mathrm{target}}$, that is, with probability at least R_{target} the available capacity clears the demand with necessity at least λ.

Proposition 3 (Conditional monotonicity in α). *For a fixed γ, if $\frac{d}{d\alpha} W_S^{\alpha}(\gamma) \leq 0$ and $\frac{d}{d\alpha} k^{\alpha}(\gamma) \geq 0$ for every subset S of the installed components, the subsets accepted at (α, γ) are nested in α and $R_{\mathrm{sys}}^{1,\gamma} \leq R_{\mathrm{sys}}^{\alpha,\gamma} \leq R_{\mathrm{sys}}^{0,\gamma}$ for $\alpha \in [0, 1]$.*

For symmetric spreads and a symmetric threshold the sign conditions hold for $\gamma \geq 1/2$; for $\gamma < 1/2$, which includes the necessity column, the ordering is reversed, and at $\gamma = 1/2$ the reliability does not depend on α. For asymmetric data neither ordering need hold (Supplementary Material S10). The solver does not rely on monotonicity; it only simplifies the interpretation of the results.

3 Design problem and exact solution

3.1 Redundancy-allocation model

Each subsystem i is assigned exactly one alternative x_i from a discrete catalog M_i, which may contain a null option (no unit installed). Option $j \in M_i$ has a fuzzy capacity $\tilde{w}_{i,j}$, a reliability $p_{i,j}$ and a capital cost $c_{i,j}$. For a declared posture (α, γ) the problem is to minimize the capital cost subject to the reliability target, and to break ties in favor of the most reliable design:

$$\min_{\mathbf{x}} \quad Z(\mathbf{x}) \;=\; \sum_{i=1}^{n} c_i(x_i) \tag{14}$$

$$\text{subject to} \quad R_{\mathrm{sys}}^{\alpha,\gamma}(\mathbf{x}) \;\geq\; R_{\mathrm{target}}, \quad x_i \in M_i,\; i = 1, \ldots, n, \tag{15}$$

$$\max_{\mathbf{x} \in \mathcal{X}^*} \quad R_{\mathrm{sys}}^{\alpha,\gamma}(\mathbf{x}), \tag{16}$$

where $\mathcal{X}^*$ is the set of minimum-cost feasible designs. When R_{target} is a contractual specification, this lexicographic formulation uses only the problem data, whereas a fuzzy-goal aggregation would add tolerance parameters[52]; it also admits a certificate of global optimality.

3.2 Reduction of the policy square

The dual-fuzzy formulation asks for one optimization per point of the square $[0, 1]^2$, and computational studies of fuzzy allocation typically report results on a coarse grid. Assume, as in the base case of Section 4, that every capacity is fuzzified with the same proportional spreads, $\tilde{w}_i = \big(w_i(1 - \delta_l),\, w_i,\, w_i(1 + \delta_r)\big)$, $\delta_l, \delta_r \in [0, 1)$.

Theorem 1 (Reduction to one threshold). *Let the capacities be proportionally fuzzified with common spreads $\delta_l, \delta_r \in [0, 1)$, let the threshold $\tilde{k}$ be an arbitrary TFN, and let $\alpha, \gamma \in [0, 1]$. With the capacity multiplier*

$$m(\alpha, \gamma) \;=\; 1 - \delta_l(1 - \alpha) + \gamma(\delta_l + \delta_r)(1 - \alpha) \;>\; 0, \tag{17}$$

which does not depend on the subset S, the success condition of Equation (4) *is equivalent to a comparison of the modal total with one scalar,*

$$W_S^{\alpha}(\gamma) \geq k^{\alpha}(\gamma) \iff W_S \;\geq\; \kappa(\alpha, \gamma) \;=\; \frac{k^{\alpha}(\gamma)}{m(\alpha, \gamma)}. \tag{18}$$

Proof. Every capacity is a scalar multiple of one common unit, $\tilde{w}_i = w_i\tilde{u}$ with $\tilde{u} = (1-\delta_l, 1, 1+\delta_r)$. Scalar multiplication distributes over the extension-principle sum[53], so $\tilde{W}_S = W_S\tilde{u}$, $W^{\alpha}_{S,L} = W_S(1-\delta_l(1-\alpha))$ and $W^{\alpha}_{S,U} = W_S(1+\delta_r(1-\alpha))$. Substituting into Equation (2) gives $W^{\alpha}_{S}(\gamma) = W_S\, m(\alpha,\gamma)$, and $m > 0$ because $\delta_l < 1$. □

The argument is the distributivity used in reductions of possibilistic constraints[20,54,55]; what it buys here comes from the weighted structure. With integer modal capacities, the pair (α,γ) enters the design problem only through the integer $T = \lceil \kappa(\alpha,\gamma) \rceil$, so two postures with the same T have the same optimal design, and the square splits into finitely many regions with closed-form boundaries. This extends parametric analysis[56] to discrete multi-choice catalogs.

Corollary 2 (Closed-form region boundaries). *For each integer threshold T and cut level $\alpha \in [0,1)$, the curve separating $\kappa(\alpha,\gamma) \le T$ from $\kappa(\alpha,\gamma) > T$ is*

$$\gamma_T(\alpha) \;=\; \frac{k^{\alpha}_U - T\big(1-\delta_l(1-\alpha)\big)}{(k^{\alpha}_U - k^{\alpha}_L) + T(\delta_l+\delta_r)(1-\alpha)}, \tag{19}$$

and, since κ decreases in γ for $\alpha < 1$, $\lceil \kappa \rceil = T$ on $\{\gamma \in [0,1] : \gamma_T(\alpha) \le \gamma < \gamma_{T-1}(\alpha)\}$.

The proof equates $\kappa = T$, which is linear in γ; at $\alpha = 1$ all TFNs collapse to their modal values and $\kappa(1,\gamma) = k_2$. Solving one crisp problem per attainable value of T therefore yields the optimal design at *every* point of the square, including the points between the grid lines. The reduction also shrinks the computation: the reliability recursion of Section 3.3 carries a state space proportional to the integer threshold, and T replaces a much larger threshold obtained by scaling the fuzzy endpoints to integers. The reduction requires common proportional spreads; with component-specific spreads the multiplier depends on the subset and each cell must be solved separately (Section 5.5).

3.3 Two-phase exact solver

Metaheuristics dominate recent allocation studies[57], but instances of moderate size remain within reach of exact search[2,58]. The solver combines a depth-first search over the catalogs with the capped-capacity convolution for weighted systems[1,4,59]. A partial design with subsystems $1,\ldots,i-1$ assigned is described by its cost Z_{part}, its capacity W_{part} and

the probability mass function π_{part} of its capacity, capped at the threshold; all capacities are the effective capacities $w^{\alpha}(\gamma)$ of the cell being solved. When option j is considered for subsystem i, $Z_{\mathrm{next}} = Z_{\mathrm{part}} + c_{i,j}$ and $q_{\mathrm{next}} = \max\{0,\, k^{\alpha}(\gamma) - W_{\mathrm{part}} - w_{i,j}\}$ is the capacity still required. Two suffix quantities bound the completions: $W_i^{\max} = \sum_{l>i} w_l^{\max}$, and $Z_i^{\mathrm{suf}}(q)$, the minimum additional cost of subsystems $i+1, \ldots, n$ that supply at least q further units of capacity, computed once by a suffix dynamic program.

Solving both lexicographic levels in one traversal would require tracking equal-cost subtrees while the cost incumbent is still decreasing. The search is therefore split into two exact phases. Phase 1 certifies the minimum feasible cost Z^*; Phase 2 fixes Z^* and searches only the designs of cost Z^* for the most reliable one. Both phases explore options in ascending cost order and apply two pruning tests: a subtree is discarded if even the largest units of the remaining subsystems cannot reach the threshold or if its cost lower bound $Z_{\mathrm{next}} + Z_i^{\mathrm{suf}}(q_{\mathrm{next}})$ *strictly* exceeds the incumbent; and if an optimistic completion, in which every unassigned subsystem carries the largest reliability and the largest effective capacity of its catalog, falls below R_{target}. The incumbent is seeded with the cost of a greedy feasible design. Algorithm 1 states the search. Capacity comparisons are exact, because the α-cut endpoints and Hurwicz values are computed in rational arithmetic and scaled to integers; reliability comparisons use the tolerance $\epsilon = 10^{-12}$, so the optimality guarantee holds up to ϵ on the reliability side.

Global optimality requires the pruning tests to be admissible. The proofs of the next three results are given in Supplementary Material S10.

Theorem 2 (Residual-capacity and suffix-cost bounds). *Let capacities and costs be non-negative, $R_{\mathrm{target}} > 0$, and Z^* the incumbent cost. (i) If $W_{\mathrm{part}} + w_{i,j} + W_i^{\max} < k^{\alpha}(\gamma)$, every completion has zero reliability. (ii) If $Z_{\mathrm{next}} + Z_i^{\mathrm{suf}}(q_{\mathrm{next}}) > Z^*$, no completion improves on the incumbent under the lexicographic objective.*

The inequality in (ii) must be strict: a completion of equal cost may still improve the secondary objective, which is why Algorithm 1 tests $>$ and not $\geq$.

Theorem 3 (Optimistic reliability bound). *Let R_{ub} be the reliability obtained by convolving the capped capacity distribution of a partial design with, for each unassigned subsystem, a virtual option carrying*

Algorithm 1: Two-phase dual-fuzzy exact solver. The state variables and suffix quantities are defined in Section 3.3; the implementation applies the pruning tests on entry to the child node.

Input: Catalogs $M_1, \dots, M_n$, threshold $\tilde{k}$, target R_{target}, cell (α, γ)
Output: Lexicographically optimal design $\mathbf{x}^*$, certified minimum cost Z^*, reliability R^*

Phase 0: compute $k^{\alpha}(\gamma)$, $w^{\alpha}_{i,j}(\gamma)$, $W_i^{\max}$ and $Z_i^{\mathrm{suf}}(\cdot)$;
Phase 1: initialize Z^* with the cost of a greedy feasible design, or ∞;
Function `SearchPhase1` $(i, Z_{\mathrm{part}}, W_{\mathrm{part}}, \pi_{\mathrm{part}})$ **:**
  if $i > n$ **then**
    if $R_{\mathrm{sys}}(\pi_{\mathrm{part}}) \geq R_{\mathrm{target}}$ **then** $Z^* \leftarrow \min(Z^*, Z_{\mathrm{part}})$;
    return;
  end
  foreach *option* $j \in M_i$ *sorted by ascending cost* **do**
    $Z_{\mathrm{next}} \leftarrow Z_{\mathrm{part}} + c_{i,j}$; $q_{\mathrm{next}} \leftarrow \max\{0,\ k^{\alpha}(\gamma) - W_{\mathrm{part}} - w_{i,j}\}$;
    if $Z_{\mathrm{next}} + Z_i^{\mathrm{suf}}(q_{\mathrm{next}}) > Z^*$ ***or*** $W_{\mathrm{part}} + w_{i,j} + W_i^{\max} < k^{\alpha}(\gamma)$ **then continue**;
    $\pi_{\mathrm{next}} \leftarrow$ `Convolve`$(\pi_{\mathrm{part}}, p_{i,j}, w_{i,j})$;
    if `OptimisticReliability`$(\pi_{\mathrm{next}}, i) < R_{\mathrm{target}}$ **then continue**;
    `SearchPhase1` $(i+1, Z_{\mathrm{next}}, W_{\mathrm{part}} + w_{i,j}, \pi_{\mathrm{next}})$;
  end
`SearchPhase1` $(1, 0, 0, \pi_0)$;
if $Z^* = \infty$ **then return** Infeasible;
Phase 2: repeat the traversal with Z^* fixed and the same pruning tests, carrying the partial design; at each leaf with $Z_{\mathrm{part}} = Z^*$ and $R_{\mathrm{sys}}(\pi_{\mathrm{part}}) > R^*$ set $R^* \leftarrow R_{\mathrm{sys}}(\pi_{\mathrm{part}})$, $\mathbf{x}^* \leftarrow \mathbf{x}_{\mathrm{part}}$;
return $(\mathbf{x}^*, Z^*, R^*)$;

the largest reliability and the largest effective capacity of its catalog. No completion has reliability greater than R_{ub}.

Theorem 4 (Termination and global lexicographic optimality). *For finite n and catalogs, Algorithm 1 terminates, and at termination $(\mathbf{x}^*, Z^*, R^*)$ solves Equations* (14)–(16)*, or the algorithm correctly declares that no design meets the target.*

A solution returned by Algorithm 1 is called *certified*. A numerical trace of both phases on a three-component system is given in Supplementary Material S1. The search space has $O(|M|^n)$ designs and evaluating one design costs $O(n \cdot T)$ with the capped convolution; the bounds do not remove the exponential worst case, and their effect in practice is reported in Section 5.6.

3.4 Cost–necessity frontier

Each cell minimizes cost subject to a reliability target at one effective threshold, and the expected necessity reliability R_N does not enter that optimization. A decision maker who prices necessity directly needs a different problem: for each required level r,

$$\min_{\mathbf{x}} Z(\mathbf{x}) \quad \text{subject to} \quad R_N(\mathbf{x}) \geq r, \tag{20}$$

and, when the design must also meet the target at the modal demand, with $R_{\text{modal}}(\mathbf{x})$ the crisp reliability at k_2,

$$\min_{\mathbf{x}} Z(\mathbf{x}) \quad \text{subject to} \quad R_N(\mathbf{x}) \geq r, \quad R_{\text{modal}}(\mathbf{x}) \geq R_{\text{target}}. \tag{21}$$

By Equation (13), R_N averages over many thresholds, so the single-threshold bounds do not apply; the frontier is computed here by exhaustive enumeration.

4 Case study: IEEE RTS-79 generation fleet

4.1 Instance and protocol

The generation fleet of the IEEE Reliability Test System[42] has $n = 10$ bays that must meet a peak regional demand $\tilde{k} = (162, 180, 198)$ MW, whose spread represents the uncertainty of the peak-load forecast, with the reliability target $R_{\text{target}} = 0.95$. Each bay offers a null option and two of the six RTS-79 unit classes of Table 3, so the catalog has $3^{10} = 59{,}049$ designs; hydro units are available only at Bays 3–6, so a fleet contains at most four of them. The availability of a unit is $p = 1 - \text{FOR} = \text{MTTF}/(\text{MTTF} + \text{MTTR})$, with the forced outage rate (FOR) and the mean times to failure and to repair computed from the unit statistics of the source. RTS-79 reports operating costs only, so the capital costs are an illustrative assumption of this study. The static reliability of a fleet is the probability of no capacity shortfall, computed from the capacity outage probability distribution of its units; for the crisp model at the modal demand, $R_{\text{sys}} \geq 0.95$ is a single-snapshot loss-of-load probability of at most 0.05 at peak demand. The complete catalog is listed in Supplementary Material S6.

Capacity spreads. In the base case every capacity is a TFN with the common proportional spread $\delta = 0.10$, $\tilde{w}_i = (0.9w_i,\ w_i,\ 1.1w_i)$. The spread has an operational motivation: the output of thermal units, and

Table 3. Generating-unit classes of the case study. Capacities w and availabilities $p = 1 - \mathrm{FOR}$ are those of RTS-79[42] (its Table 4); capital costs are an illustrative assumption of this study. The last column lists the bays at which each class can be installed.

Unit class	w (MW)	p	Cost (M$)	M$/MW	Bays
Oil steam	12	0.98	14	1.17	1–2
Oil combustion turbine	20	0.90	18	0.90	1–4
Hydro	50	0.99	45	0.90	3–6
Coal steam	76	0.98	68	0.89	5–8
Oil steam	100	0.96	85	0.85	7–10
Coal steam	155	0.96	135	0.87	9–10

of combustion turbines in particular, falls in hot weather as combustion air density drops, whereas hydraulic output is largely insensitive to air temperature. RTS-79 gives seasonal capacity limits for its hydro units but no partial-outage data for thermal units, so the base case uses one spread for all units, and Section 5.5 replaces it by unit-specific spreads.

Protocol. The instance is solved on the 30-cell grid $\alpha \in \{0.0, 0.3, 0.5, 0.7, 0.9, 1.0\}$, $\gamma \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, and the complete parametric solution of Theorem 1 is computed over the whole square. Two instances derived from published allocation problems, a 14-subsystem instance based on Fyffe et al.[60] and a 10-subsystem instance based on the pharmaceutical plant of Garg and Sharma[58], are used to check whether the findings recur; neither is the original optimization problem, and their catalogs are given in Supplementary Material S4 and S5, and the replication of the findings in Supplementary Material S8. A synthetic family measures scalability (Section 5.6). The computational pipeline and the script behind each result are documented in Supplementary Material S2.

4.2 Baselines

The complete parametric solution is compared with three kinds of baseline.

Grid. On the region of the square whose optimum is at no cell of the 30-cell grid we measure how often the design of the nearest grid cell fails the reliability target, and how much more than the optimum the cheapest grid design that meets it costs.

Simpler models with the same data. The crisp model uses the modal capacities and demand, so its threshold is $\lceil k_2 \rceil$; the model with fuzzy capacities only has threshold $\lceil k_2 / m(\alpha, \gamma) \rceil$; the model with fuzzy demand

only has threshold $\lceil k^{\alpha}(\gamma)\rceil$. Each baseline design is the optimum at its own threshold and is evaluated at the threshold of the dual-fuzzy model at the same posture: it either fails the target or meets it at a cost no lower than the optimum. The probabilistic counterpart treats the same imprecision as randomness: a common derating factor $\Theta \in [1-\delta_l,\, 1+\delta_r]$ and the demand $D \in [k_1, k_3]$ are independent random variables with densities on the supports of the fuzzy data, success is $\Theta W_S \geq D$, that is $W_S \geq \hat{T} = \lceil D/\Theta \rceil$, and

$$R_{\mathrm{prob}}(\mathbf{x}) = \mathbb{E}\big[R(\mathbf{x}, \hat{T})\big] = \sum_{T} \Pr\{\hat{T} = T\}\, R(\mathbf{x}, T), \tag{22}$$

where $R(\mathbf{x}, T)$ is the crisp reliability at the integer threshold T. The probabilistic design minimizes cost subject to $R_{\mathrm{prob}}(\mathbf{x}) \geq R_{\mathrm{target}}$; we read the membership functions as triangular densities and repeat the computation with uniform densities. Because R_{prob} mixes many thresholds, it is solved by enumeration.

Algorithm. A genetic algorithm with matched computational effort (Section 5.6).

The reference of the model baselines is the declared dual-fuzzy rule: the comparison measures what a planner who regards both sides as imprecise, and has declared a posture, loses by solving a simpler model; it does not show that the simpler models are wrong for a planner who accepts their assumptions. By Theorem 1 a one-sided model with a recalibrated spread can reproduce the dual threshold in any one cell; the comparison is made at the same data, in which the capacity and demand spreads are separate inputs. All areas are integrals over the square on the closed-form regions of Corollary 2, computed by the midpoint rule in α, and shares are of the part of the square on which some design meets the target (Supplementary Material S7).

5 Results

5.1 The complete solution and what a grid misses

For the RTS-79 fleet, κ ranges over $[147.27,\, 220.00]$ MW, yielding 73 integer thresholds and the 7 distinct optimal fleets of Table 4, five of which are optimal at some cell of the 30-cell grid. Figure 1 shows this solution. In panel (a) the thin lines separate the 73 crisp subproblems, and the six thick lines are the curves $\gamma_T(\alpha)$ of Equation (19) for $T =$

150, 155, 162, 176, 200, 205, the thresholds at which the optimal fleet changes. Panel (b) plots the same solution against κ: the minimum cost is a non-decreasing step function of the effective threshold. The regions are unequal in size. The crisp-optimal fleet D5 (4×50 MW hydro, \$180 M) is optimal on 61% of the square, and D4 on a further 25%.

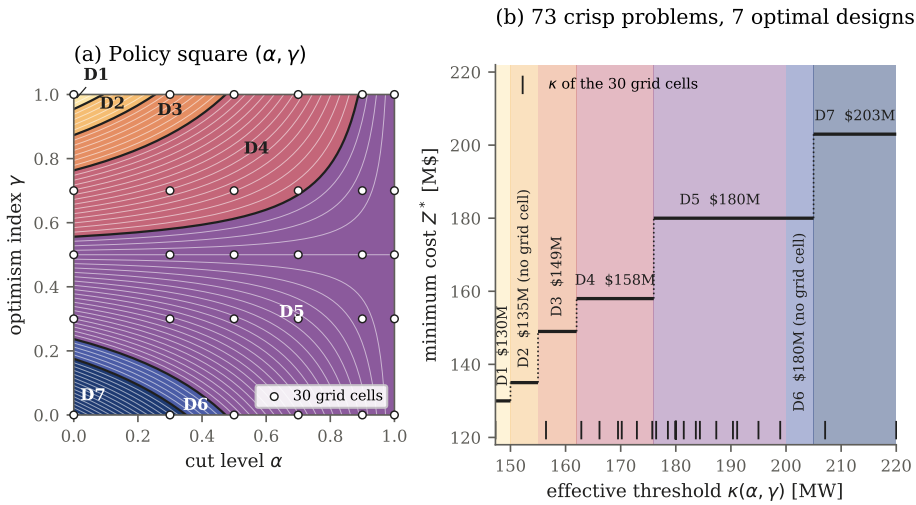


Figure 1. Complete parametric solution of the IEEE RTS-79 fleet ($\delta = 0.10$). (a) Optimal design over the policy square. Thin lines are the level curves $\kappa(\alpha, \gamma) = T$ separating the 73 crisp subproblems; thick lines are the boundaries $\gamma_T(\alpha)$ of Corollary 2 at which the optimal fleet changes; circles are the 30 grid cells (Table 5). (b) Minimum cost Z^* against the effective threshold κ, with ticks at the κ values of the grid cells. Designs D1–D7 are numbered by increasing cost; D5 and D6 cost the same. D2 and D6 are optimal on regions that contain no grid cell.

Table 4. Design menu of the IEEE RTS-79 fleet: the seven designs of the complete parametric solution ($\delta = 0.10$), with the necessity and possibility reliabilities R_N, R_Π of each design, the share of the policy square on which it is optimal and the number of the 30 grid cells that prescribe it.

Design	Fleet	Cost (M$)	T range on square	R_N	R_Π	Area (%)	Grid cells
D1	1×50 MW Hydro + 1×100 MW Oil	130	148–150	0.000	0.086	0.2	1
D2	1×155 MW Coal	135	151–155	0.000	0.244	1.6	0
D3	3×50 MW Hydro + 1×12 MW Oil	149	156–162	0.000	0.452	5.0	1
D4	2×50 MW Hydro + 1×76 MW Coal	158	163–176	0.000	0.853	25.3	6
D5	4×50 MW Hydro	180	177–200	0.506	0.964	61.1	20
D6	1×50 MW Hydro + 1×155 MW Coal	180	201–205	0.617	0.953	3.3	0
D7	3×50 MW Hydro + 1×76 MW Coal	203	206–220	0.951	0.978	3.5	2

Two of the seven designs are optimal on regions of positive area that contain none of the 30 grid cells. D2 (1×155 MW coal, \$135 M, 1.6% of the square) is optimal on the possibility column $\gamma = 1$ for $\alpha \in (0.091, 0.254]$, between grid cells that prescribe D1 and D3. D6 (1×50 MW hydro and 1×155 MW coal, \$180 M, 3.3%) is optimal on the necessity column $\gamma = 0$ for $\alpha \in [0.351, 0.474)$, between grid cells that prescribe D7 and D5. Neither neighbor is optimal there: D5 supplies 200 MW and cannot meet an effective threshold above 200 MW, and D7 costs \$23 M more than D6. The design changes discontinuously at the region boundaries, so interpolating between two grid nodes can name a design that is optimal at neither. Each of the 30 cells was re-solved independently and agreed with the region assigned to it on cost, reliability and design.

Table 4, the design menu, also shows a trade-off that the grid hides. D5, optimal on 61% of the square, and D6, optimal on 3.3% and at no grid cell, have the same cost; D6 has the higher necessity reliability (0.617 against 0.506) and D5 the higher possibility reliability (0.964 against 0.953). A decision maker who gives priority to the necessity of meeting demand would therefore choose D6, which a grid study does not offer. On the 4.8% of the square where D2 or D6 is optimal, the design of the nearest grid cell fails the target on 53% of the area, and the cheapest grid design that meets it costs up to 12.8% more than the optimum. The two derived instances show the same pattern, with 14 of 29 and 6 of 20 optimal designs at no grid cell (Supplementary Material S8).

5.2 The declared posture sets the design, its cost and its necessity

Table 5 reports the certified optimum in every cell of the grid; every cell admits a feasible fleet. We call $(0, 1)$ the optimistic boundary cell: by Proposition 1 the column $\gamma = 1$ expresses the possibility constraint $\Pi \geq \alpha$ for $\alpha > 0$, and $(0, 1)$ is its limit, at which only the supports of capacity and demand have to overlap. The minimum investment ranges from \$130 M in the optimistic boundary cell, with effective threshold $T = 148$ MW, through \$180 M in the modal cell $(0.5, 0.5)$, with $T = 180$ MW, to \$203 M in the necessity cell $(0.0, 0.0)$, with $T = 220$ MW. The optimal fleet changes with the posture:

- In the optimistic boundary cell the rule credits each unit with its upper capacity ($1.1w_i$) and compares it with the lower demand bound

Table 5. Exact global optimum over the (α, γ) grid, IEEE RTS-79-parameterized generation fleet case study. T is the effective integer threshold of Theorem 1 in MW; R_{sys} is the reliability of the prescribed design under the Hurwicz rule at that cell, and R_N, R_Π are the graded necessity and possibility reliabilities of the same design (all 30 cells admit feasible certified optima).

α	γ	T	Cost (M$)	R_{sys}	R_N	R_Π	Nodes	Time (s)
0.0	0.0	220	203	0.9509	0.951	0.978	8,366	0.00
0.3	0.0	208	203	0.9509	0.951	0.978	17,360	0.01
0.5	0.0	199	180	0.9606	0.506	0.964	1,919	0.00
0.7	0.0	192	180	0.9606	0.506	0.964	5,573	0.00
0.9	0.0	184	180	0.9606	0.506	0.964	10,379	0.01
1.0	0.0	180	180	0.9606	0.506	0.964	12,185	0.01
0.0	0.3	195	180	0.9606	0.506	0.964	4,337	0.00
0.3	0.3	191	180	0.9606	0.506	0.964	5,573	0.00
0.5	0.3	188	180	0.9606	0.506	0.964	8,525	0.01
0.7	0.3	185	180	0.9606	0.506	0.964	9,587	0.01
0.9	0.3	182	180	0.9606	0.506	0.964	12,083	0.01
1.0	0.3	180	180	0.9606	0.506	0.964	12,185	0.01
0.0	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
0.3	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
0.5	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
0.7	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
0.9	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
1.0	0.5	180	180	0.9606	0.506	0.964	12,185	0.01
0.0	0.7	167	158	0.9605	0.000	0.853	3,458	0.00
0.3	0.7	171	158	0.9605	0.000	0.853	2,102	0.00
0.5	0.7	173	158	0.9605	0.000	0.853	1,346	0.00
0.7	0.7	176	158	0.9605	0.000	0.853	866	0.00
0.9	0.7	179	180	0.9606	0.506	0.964	12,221	0.01
1.0	0.7	180	180	0.9606	0.506	0.964	12,185	0.01
0.0	1.0	148	130	0.9504	0.000	0.086	329	0.00
0.3	1.0	157	149	0.9509	0.000	0.452	6,725	0.00
0.5	1.0	163	158	0.9605	0.000	0.853	5,438	0.00
0.7	1.0	170	158	0.9605	0.000	0.853	3,362	0.00
0.9	1.0	177	180	0.9606	0.506	0.964	13,241	0.01
1.0	1.0	180	180	0.9606	0.506	0.964	12,185	0.01

(162 MW). The target is met by one 50 MW hydro unit and one 100 MW oil unit (150 MW modal capacity, $R_{\mathrm{sys}} = 0.9504$, $R_N = 0.0000$), the remaining bays left empty.

- In the modal cell the optimal fleet is four 50 MW hydro units (200 MW modal capacity), with $R_{\mathrm{sys}} = 0.9606$ and $R_N = 0.5056$.
- In the necessity cell the capacity of the working units must clear the upper demand bound (198 MW) even at the lower end of the common spread ($0.9w_i$). The solver commits three 50 MW hydro

units and one 76 MW coal unit (226 MW modal capacity), with $R_N = R_{\mathrm{sys}} = 0.9509$.

The optimal fleets favor the hydro ($p = 0.990$) and 76 MW coal ($p = 0.980$) units, whose cost per MW (0.90 and 0.89 M\$) is close to that of the larger and less available 100 MW oil and 155 MW coal units ($p = 0.960$; 0.85 and 0.87 M\$). The solver solves every cell in ≤ 0.01 seconds, exploring between 329 and $17{,}360$ nodes.

The pair (α, γ) thus specifies a risk posture, and the solver reports, in monetary units, the rate at which capital expenditure trades against the strength of the guarantee. All intervals collapse at $\alpha = 1$, and every column gives the crisp cost of \$180 M. As α decreases the columns separate: under the pessimistic rule ($\gamma = 0$) the cost rises to \$203 M ($+12.8\%$), and under the optimistic rule ($\gamma = 1$) it falls to \$158 M at $\alpha = 0.7$ and to \$130 M at $\alpha = 0$ (-27.8%). The savings come with a weaker guarantee: a design prescribed on the column $\gamma = 1$ is one whose success is possible at the stated confidence, not necessary, and on the design menu every design cheaper than \$180 M has $R_N = 0$. At $\alpha = 0$ the necessity reliability of the prescribed design falls from 0.95 at $(0, 0)$ to 0.00 at $(0, 1)$ (Supplementary Material S6 plots both quantities against γ).

Crisp and dual-fuzzy reliability of the same design. Table 6 compares, for each design D1–D7, the classical crisp reliability with the reliability under the dual-fuzzy rule, and separates the contribution of each fuzzified side. First, the crisp model assigns zero reliability to D1–D4, whose modal capacity is below the modal demand of 180 MW; in the optimistic boundary cell the same fleets reach 0.950–0.970. Second, part of the effect appears only when both sides are fuzzy. For D5 and D6, fuzzifying either the capacities or the demand alone leaves the crisp reliability unchanged in the necessity cell (0.9606 and 0.9504): for D5, $0.9 \times 200 = 180$ MW still covers the modal demand, and 200 MW still covers the upper demand of 198 MW. Fuzzifying both sides reduces the reliability of both designs to zero, since $0.9 \times 200 = 180$ MW and $0.9 \times 205 = 184.5$ MW fall short of 198 MW. Third, the necessity design D7 keeps its crisp reliability (0.9509) in the necessity cell under every variant. A reliability analysis that fuzzifies one side only would therefore report D5 and D6 as adequate in the necessity cell, although neither is.

Table 6. Crisp and dual-fuzzy reliability of the same designs (IEEE RTS-79 fleet, designs D1–D7 of Figure 1). "Crisp": modal capacities against the modal demand of 180 MW. In the necessity cell $(0,0)$ and the optimistic boundary cell $(0,1)$, "$\tilde{w}$ only" fuzzifies the capacities and keeps the demand at 180 MW, "$\tilde{k}$ only" fuzzifies the demand and keeps the modal capacities, and "both" is the dual-fuzzy model.

				Necessity cell $(0,0)$			Optimistic boundary cell $(0,1)$		
Design	MW	Cost (M$)	Crisp	$\tilde{w}$ only	$\tilde{k}$ only	both	$\tilde{w}$ only	$\tilde{k}$ only	both
D1	150	130	0	0	0	0	0	0	0.9504
D2	155	135	0	0	0	0	0	0	0.9600
D3	162	149	0	0	0	0	0	0.9509	0.9703
D4	176	158	0	0	0	0	0.9605	0.9605	0.9605
D5	200	180	0.9606	0.9606	0.9606	0	0.9606	0.9606	0.9994
D6	205	180	0.9504	0.9504	0.9504	0	0.9504	0.9504	0.9600
D7	226	203	0.9509	0.9509	0.9509	0.9509	0.9797	0.9797	0.9991

Table 7. Model baselines evaluated under the declared dual-fuzzy rule. Each baseline uses the same data and the same posture (α,γ) but a simpler model; its design is evaluated at the effective threshold of the dual-fuzzy model. Shares are of the part of the policy square on which some design meets the reliability target. "Fails": the baseline design does not meet the target; "Costlier": it meets the target at a cost above the optimum; "Max. excess": largest relative cost excess where it meets the target. Probabilistic counterpart: triangular distributions on the supports of the fuzzy data (Section 4.2).

	IEEE RTS-79			Fyffe-derived			Pharmaceutical-derived		
Baseline	Fails (%)	Costlier (%)	Max. excess (%)	Fails (%)	Costlier (%)	Max. excess (%)	Fails (%)	Costlier (%)	Max. excess (%)
Crisp (modal data)	6.8	32.1	38.5	50.0	37.6	64.3	49.9	33.4	33.7
Fuzzy capacities only	6.8	17.1	21.5	37.7	37.6	42.9	33.5	33.4	25.0
Fuzzy demand only	6.8	16.7	21.5	33.0	27.7	21.4	30.9	30.3	16.3
Probabilistic counterpart	6.8	93.2	49.2	–	–	–	33.5	66.5	44.6

5.3 Ignoring a source of imprecision is costly

Table 7 evaluates the model baselines under the declared rule. The crisp design fails the reliability target on 6.8%, 50.0% and 49.9% of the square of the three instances and, where it meets the target, costs more than the optimum on a further 32.1%, 37.6% and 33.4%, by up to 38.5%, 64.3% and 33.7%. Fuzzifying one side reduces both effects without removing them: with fuzzy capacities only the design fails on 6.8%, 37.7% and 33.5% of the square, and with fuzzy demand only on 6.8%, 33.0% and 30.9%. On the RTS-79 fleet the three model baselines fail on the same region, the part of the necessity side on which the dual threshold exceeds 200 MW. No one-sided threshold reaches it, since $k_2/m(\alpha,\gamma) \le 200$ MW and $k^{\alpha}(\gamma) \le 198$ MW, and there all three prescribe D5, which supplies

200 MW. The probabilistic counterpart is not computed for the Fyffe-derived instance, whose 4^{14} designs cannot be enumerated. Supplementary Material S7 gives these results in full.

The probabilistic counterpart behaves differently. With triangular densities it selects four 50 MW hydro units and one 12 MW oil unit at \$194 M ($R_{\mathrm{prob}} = 0.9605$). The probability that the effective threshold exceeds 200 MW is 0.034, and the added oil unit raises the reliability at the thresholds from 201 to 212 MW from zero to 0.941: enough for the average of Equation (22), but below the target at each of those thresholds. Under the declared rule the design therefore fails at every posture whose threshold exceeds 200 MW, and at every other posture a cheaper design is optimal. It meets crisp adequacy ($R_{\mathrm{modal}} = 0.961$), but its expected necessity reliability, $R_N = 0.779$, is below the 0.880 that the adequacy-constrained cost–necessity frontier offers at the same budget. With uniform densities on the same supports the probabilistic design changes to four hydro units and one 20 MW combustion turbine at \$198 M, so the prescription depends on a distributional assumption that the data do not fix. On the Pharmaceutical-derived instance the probabilistic design (cost 133) is likewise optimal at no posture and fails the target on 33.5% of the square.

5.4 Menu designs need not be efficient in necessity

Each menu design minimizes cost subject to the target in its own cell; it is not selected for its necessity reliability. We enumerated all $3^{10} = 59{,}049$ designs and solved the frontier problems of Section 3.4 (Figure 2). Without the adequacy constraint the frontier contains designs cheaper than \$180 M with positive necessity, such as the \$170 M fleet of two 100 MW oil units, whose crisp adequacy is only $0.960^2 = 0.9216$. With the adequacy constraint no design cheaper than \$180 M is admissible, and between D6 and D7 the frontier contains three designs that no Hurwicz cell prescribes: a planner bound by adequacy obtains $R_N = 0.6392$ for \$190 M, $R_N = 0.8804$ for \$194 M and $R_N = 0.9171$ for \$198 M, against 0.9509 for D7 at \$203 M. The menu and the frontier differ because each menu design is the cheapest design at a single effective threshold, and R_N does not enter that optimization. Supplementary Material S6 lists the frontier designs and the dominated catalog fleets.

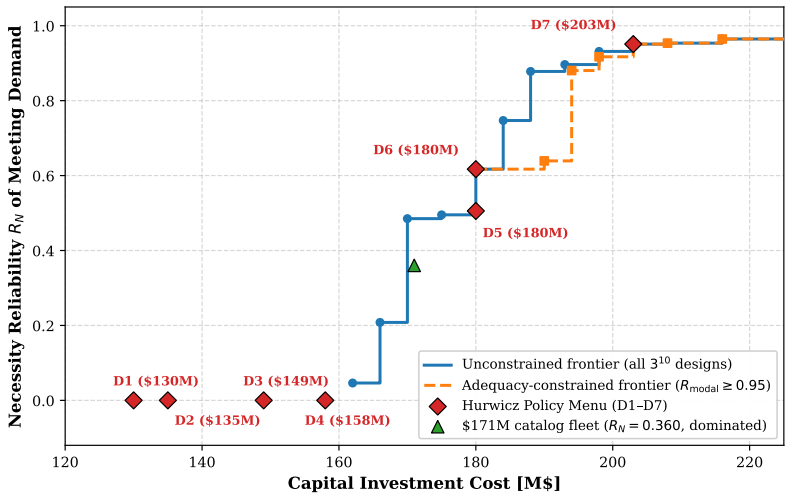


Figure 2. Cost–necessity frontiers of the IEEE RTS-79 catalog (exhaustive enumeration of all $3^{10} = 59{,}049$ designs). Solid line: minimum cost subject to $R_N \geq r$ (Equation (20)). Dashed line: the same with crisp adequacy $R_{\text{modal}} \geq 0.95$ (Equation (21)). Diamonds: design menu D1–D7; D5 and D6 both cost \$180 M. Triangle: the \$171 M catalog fleet with the largest R_N (0.360), dominated by the \$170 M frontier design.

5.5 *Robustness to the spread model and the target*

The capacity spread, the demand spread and the target are assumptions rather than data, and Supplementary Material S6 varies each of them on the fleet. The main results are as follows.

- *Size of the common spread.* For δ from 0.02 to 0.20 the necessity-cell threshold $T = \lceil 198/(1-\delta) \rceil$ rises from 203 to 248 MW and the cost from \$180 M to \$265 M ($+47.2\%$); the modal cell stays at $T = 180$ MW and \$180 M for every δ, because $m(\alpha, 1/2) = 1$ for symmetric spreads; in the optimistic boundary cell the cost falls to \$127 M at $\delta = 0.20$, with zero necessity of meeting demand.
- *Asymmetric spreads.* Seasonal regimes such as summer, $(\delta_l, \delta_r) = (0.15, 0.05)$, raise the necessity-cell cost to \$231 M at $T = 233$ MW. Theorem 1 allows $\delta_l \neq \delta_r$, so every regime is solved with the same certificate.
- *Demand forecast.* Widening the demand half-width from 0.02 to 0.20 raises the necessity-cell cost from \$180 M to \$256 M and lowers the optimistic boundary cost to \$127 M.

- *Reliability target.* At $R_{\text{target}} = 0.95$ the cost-minimizing fleets of the three cells are series arrangements at their threshold: the loss of any single unit drops the working capacity below T. Over $R_{\text{target}} \in \{0.90, \ldots, 0.99\}$, a fleet that still reaches T after the loss of its largest unit becomes optimal at $R_{\text{target}} = 0.97$ in the modal cell ($212 M) and at 0.98 in the other two ($180 M and $266 M).
- *Unit-specific spreads.* With the hydro capacity $(45, 50, 50)$ MW given by the seasonal limits of RTS-79 and assumed thermal spreads, the reduction fails and each of the 30 cells is solved by the unreduced exact search (all within 600 s). The optimal fleet changes in 7 of the 30 cells and its cost in 6, in both directions: on the necessity side the common spread over-invests (at $(0.3, 0)$ the unit-specific optimum costs $180 M instead of $203 M), and on the possibility side it under-invests by $9–22 M in five cells. Under perturbed thermal spreads one cell selects a $171 M fleet (one 155 MW coal unit and two 20 MW combustion turbines) that is the crisp optimum of no integer threshold, so heterogeneous spreads can select a design that only the unreduced search finds.

Across these variants the declared posture keeps changing the prescribed fleet and its cost, in both directions.

5.6 Computational performance

On a synthetic family with three choices per subsystem, $n \in \{8, 10, \ldots, 20\}$ and three demand profiles, all 21 instances were certified within a 600 s budget; the slowest, $n = 20$ under the design profile, took 459.24 s and evaluated the exact reliability of only 89 of 3.49×10^9 complete designs. Re-solving the $n = 14$ instance without the reduction, against a threshold obtained by scaling the fuzzy endpoints to integers ($16{,}767$ against $T = 66$), took 60.69 s against 0.268 s, a factor of 226.8, close to the ratio of the thresholds; the reduced and unreduced formulations agreed on all 12 instances up to $n = 14$. A genetic algorithm given the same computing budget and an equal-cost local refinement reached the minimum cost on all 17 instances tested, but at $n = 20$ some seeds settled on a design of the same cost whose reliability is 0.9035 against the certified 0.9112. Without a certificate, nothing in the heuristic's output indicates that a more reliable design exists at the same cost.

Supplementary Material S9 gives the scaling table, the ablation and the genetic-algorithm protocol.

6 Discussion

6.1 Choosing the posture

The square divides into three zones. Low α with low γ compares the worst-case capacity with the worst-case threshold on the widest cut; by Proposition 1 the cell $(\alpha, 0)$ demands necessity $1 - \alpha$. This zone gives the highest cost (\$203 M on the fleet) and suits safety-critical infrastructure in which shortfalls carry severe penalties. Low α with high γ approaches the optimistic boundary cell, with the lowest cost (\$130 M) and zero necessity of meeting demand; it suits non-critical deployments in which occasional shortfalls are tolerable. As $\alpha \to 1$ all columns converge on the crisp design (\$180 M). The pair (α, γ) need not be chosen as two free numbers; there are three ways of fixing it, in decreasing order of what they require from the decision maker.

(i) A prescribed necessity level. When a regulation, a contract or an internal standard requires demand to be met with necessity at least λ, Proposition 1 fixes the cell $(1 - \lambda, 0)$, and no optimism index has to be elicited. A possibility or credibility requirement fixes a cell in the same way (Table 2).

(ii) A design menu. Without such a requirement, the decision maker can choose among designs instead of parameters. The complete parametric solution is a finite menu: on the RTS-79 fleet the seven designs of Table 4, each with its cost, its necessity and possibility reliabilities and the share of the square on which it is optimal. A capital budget $\bar{Z}$ is a special case: the menu gives the design with the largest necessity reliability among those costing at most $\bar{Z}$. With $\bar{Z}$ set to \$150 M no affordable design has a positive necessity of meeting demand, \$180 M buys $R_N = 0.617$ (D6, the better of the two \$180 M designs) and \$203 M buys $R_N = 0.951$; under crisp adequacy the catalog offers $R_N = 0.880$ for \$194 M and 0.917 for \$198 M (Section 5.4), designs that are not on the menu.

(iii) An elicited optimism index. If an interior γ is wanted, it follows from Equation (2) once the decision maker states a certainty equivalent C^* for an interval $[C_{\min}, C_{\max}]$: $\gamma = (C^* - C_{\min})/(C_{\max} - C_{\min})$. Because the optimal design is piecewise constant in γ, the elicited value matters only through the region it falls in, which Corollary 2 gives exactly. At

$\alpha = 0.5$, for example, the modal design D5 is optimal for every $\gamma < 0.612$, so an elicitation error below that value does not change the prescription; at $\alpha = 0$ the necessity design D7 is optimal only for $\gamma < 0.175$. The attitude of an individual manager is a weaker basis for safety-critical infrastructure than a prescribed necessity level, and we regard (iii) as a fallback.

What the baselines imply. A planner who solves the crisp model, or fuzzifies one side only, commits without stating it to a design that is optimal for some postures and not for others, and Section 5.3 shows how large the part of the square is on which that design fails the target. The probabilistic counterpart is not a neutral alternative: it needs densities that expert-based data do not fix, and on the case study its prescription changed with the density. Declaring the posture and solving over the whole square makes the commitment explicit and its consequences measurable. Possibilistic evaluation reduces to comparisons of α-cut endpoints, which is what allows the exact solver to propagate capped capacity distributions; Liu's uncertainty theory[61] is a different formalism, and replacing the possibilistic oracle by the uncertain-measure oracle of Wei et al.[26] changes the prescription on both derived instances (Supplementary Material S4; the oracle is verified in Supplementary Material S3).

6.2 Limitations

1. Certified solution is demonstrated up to $n = 20$ with three choices per subsystem; larger systems may require decomposition or hybridization with heuristics.
2. The reduction of Theorem 1 requires capacity spreads that are proportional and common to all components ($\delta_l \neq \delta_r$ is allowed). With component-specific spreads the exact search still applies cell by cell, at a higher computational cost, and the crisp family of designs need not contain the optimum.
3. The cost–necessity frontier and the probabilistic counterpart average over many thresholds, so they were computed by exhaustive enumeration, which limits them to catalogs of moderate size. Robust counterparts, such as budgeted uncertainty[62] or minimax regret[41], are not points of the square and were not computed.
4. Components are statistically independent; shared-load and cascading-failure dependencies[23] are not modeled.

5. The capital costs of the case study are an assumption, the two derived instances are adaptations rather than the original problems, and the results depend on the shapes of the membership functions, which should be elicited and reviewed with domain experts before industrial use.

7 Conclusion

When the component capacities and the demand threshold of a weighted k-out-of-n:G system are both imprecise, the success condition needs a comparison rule. This paper declares that rule as a Hurwicz comparison on α-cuts, which contains the crisp model, the interval worst and best cases and the necessity-, possibility- and credibility-based chance constraints as points of one policy square. Under common proportional capacity spreads the square acts on the design problem through one effective integer threshold, so the optimal design is known at every point of the square from a number of crisp problems of the same order as a 30-cell grid, each certified by a two-phase branch-and-bound algorithm. On a generation fleet built from the IEEE Reliability Test System the complete solution is a menu of seven fleets costing \$130–203 M; a 30-cell grid over the indices never finds two of them, including a \$180 M fleet with a higher necessity of meeting demand than the crisp-optimal fleet of the same cost, and the crisp design fails the target or overpays on a large part of the square. Simpler models of the imprecision thus commit the planner to an undeclared posture, whereas the complete solution prices each posture and lets the planner choose a design. Extending the reduction to component-specific spreads, an exact method for the cost–necessity frontier and dependence between components are the main directions for further work.

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Statements and Declarations

Ethical considerations

Not applicable. The study involves no human participants, human data or animals.

668 ### Consent to participate

669 Not applicable.

670 ### Consent for publication

671 Not applicable.

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678 ### Data availability

679 All source code, benchmark data and computational artifacts required
680 to reproduce the reported results are publicly available at `https://`
681 `github.com/k-sarikaya/FuzzyWeighted`. The case study and
682 the two derived instances are built from published sources[42,58,60]; no
683 proprietary or restricted data were used. The script that produces each
684 reported result is listed in Supplementary Material S2.

685 ### Use of generative AI

686 During the preparation of this work the author used generative AI tools to
687 assist with language editing. After using these tools the author reviewed
688 and edited the content as needed and takes full responsibility for the
689 content of the published article.

690 ### Supplemental material

691 Supplemental material for this article is available online.

692 ## References

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